

		Course Name	Power Plant Electrical Equipment (MPA-ELE 25/26Z)		
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		Year	2025	Study group	1
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Task Number	Task Name				
C	Testing Protection against Unbalanced Loading				

Assignment

The objectives of this laboratory exercise were:

1. **Verify Principle:** To understand the principle of generator protection against unbalanced loading (Negative Sequence Protection).
2. **Configuration:** To configure the test setup using Omicron software and hardware.
3. **Measurements:** To perform time-trip measurements for different unbalanced current injections (I_2) using specific K factors and Start Values.
4. **Analysis:** To calculate the theoretical trip times based on the negative sequence formula and compare them with the measured results using graphical analysis.

1 Theoretical Introduction

Unbalanced loading in a generator creates negative sequence currents (I_2) in the stator. These currents produce a magnetic field rotating opposite to the rotor, inducing double-frequency currents in the rotor body. This leads to excessive heating (K) and potential damage.

1.1 Calculation of Negative Sequence Current

The negative sequence current I_2 is calculated from the phase currents (I_a, I_b, I_c) using the symmetrical component formula:

$$I_2 = \frac{1}{3} |I_a + a^2 I_b + a I_c| \quad (1)$$

Where $a = 1 \angle 120^\circ$.

Test Configuration: In this specific laboratory session, although the physical wiring connected all three phases, the test software was configured to inject current only into phases L2 (I_b) and L3 (I_c) with equal magnitudes (I_{inj}). Phase L1 was inactive (0 A).

$$I_a = 0, \quad I_b = I_{inj}, \quad I_c = I_{inj}$$

Substituting these values (assuming 0° angle shift for test inputs) into the symmetrical component formula:

$$I_2 = \frac{1}{3} |0 + a^2(I_{inj}) + a(I_{inj})| = \frac{I_{inj}}{3} |a^2 + a|$$

Since $|a^2 + a| = 1$, the relationship simplifies to:

$$I_2 = \frac{I_{injected}}{3} \quad (2)$$

1.2 Trip Characteristic

The protection relay operates based on the heating constant K (Machine Time) and the Start Value (Threshold). The theoretical trip time (t) is calculated as:

$$t = \frac{K}{\left(\frac{I_2}{I_N}\right)^2 - (\text{StartValue})^2} \quad (3)$$

Where $I_N = 1$ A (Nominal Current).

2 Procedure and Wiring

2.1 Setup

The protection was tested using an **Omicron 363** advanced protection test set. The software used to control the injection was the Omicron Test Universe.

2.2 Wiring Schema

The wiring involved connecting the current outputs of the Omicron to the measuring inputs of the protection module (AIM4/REM 543). The inputs used were 7, 9, and 11, with returns on 8, 10, and 12.

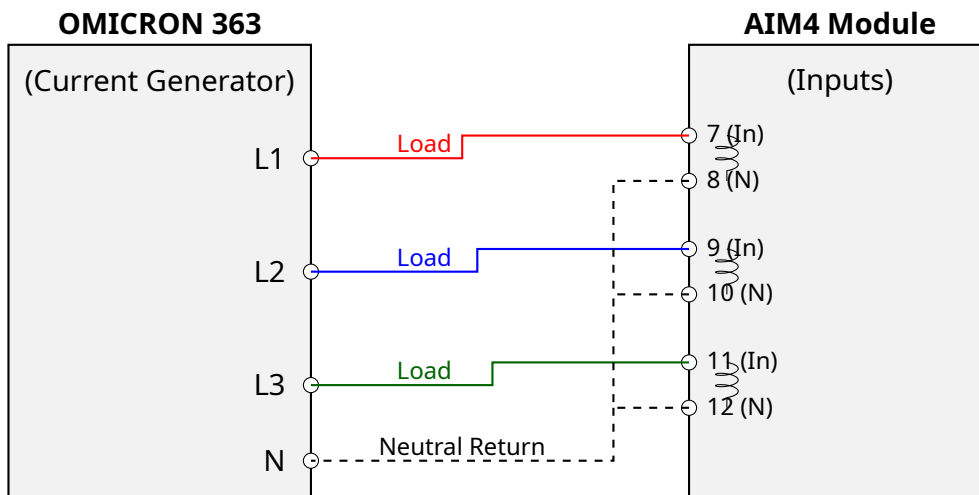


Figure 1: Schematic representation of the physical wiring. All three phases were connected to inputs 7, 9, and 11. However, during the test, current was injected only into L2 and L3 via software.

3 Measurement Results

The following tables present the measured trip times compared to the theoretical calculated times. **Note:** I_2 (Calculated) is derived as $I_{injected}/3$ due to the L2+L3 injection method. $I_N = 1A$.

3.1 Measurement 1

Settings: Machine Time (K) = 28 s, Start Value = 0.05.

Step	$I_{L2} = I_{L3}$ (A)	I_2 (Calc) [A]	$T_{measured}$ (s)	$T_{calculated}$ (s)
1	1.0	0.333	260.60	258.98
2	1.5	0.500	114.10	113.13
3	2.0	0.667	63.82	63.35
4	2.5	0.833	40.78	40.46
5	3.0	1.000	28.26	28.07
6	3.5	1.167	20.78	20.64
7	4.0	1.333	15.89	15.77

Table 1: Data for Measurement 1 ($K = 28$).

3.2 Measurement 2

Settings: Machine Time (K) = 36 s, Start Value = 0.05.

Step	$I_{L2} = I_{L3}$ (A)	I_2 (Calc) [A]	$T_{measured}$ (s)	$T_{calculated}$ (s)
1	1.0	0.333	335.00	332.97
2	1.5	0.500	146.60	145.45
3	2.0	0.667	82.04	81.45
4	2.5	0.833	52.42	52.02
5	3.0	1.000	36.33	36.09
6	3.5	1.167	26.60	26.54
7	4.0	1.333	20.42	20.28

Table 2: Data for Measurement 2 ($K = 36$).

3.3 Measurement 3

Settings: Machine Time (K) = 28 s, Start Value = 0.05. (Repeat Test).

Step	$I_{L2} = I_{L3}$ (A)	I_2 (Calc) [A]	$T_{measured}$ (s)	$T_{calculated}$ (s)
1	1.0	0.333	269.80	258.98
2	1.5	0.500	115.80	113.13
3	2.0	0.667	64.35	63.35
4	2.5	0.833	40.98	40.46
5	3.0	1.000	28.36	28.07
6	3.5	1.167	20.81	20.64
7	4.0	1.333	15.92	15.77

Table 3: Data for Measurement 3 ($K = 28$).

4 Diagrams and Analysis

1. Measured vs Calculated (Measurement 1, K=28)

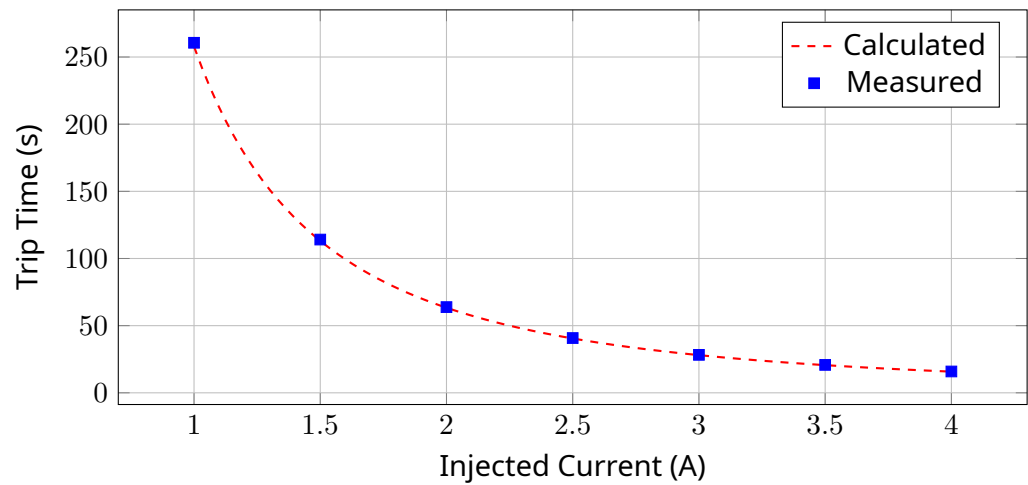


Figure 2: Results for Measurement 1.

2. Measured vs Calculated (Measurement 2, K=36)

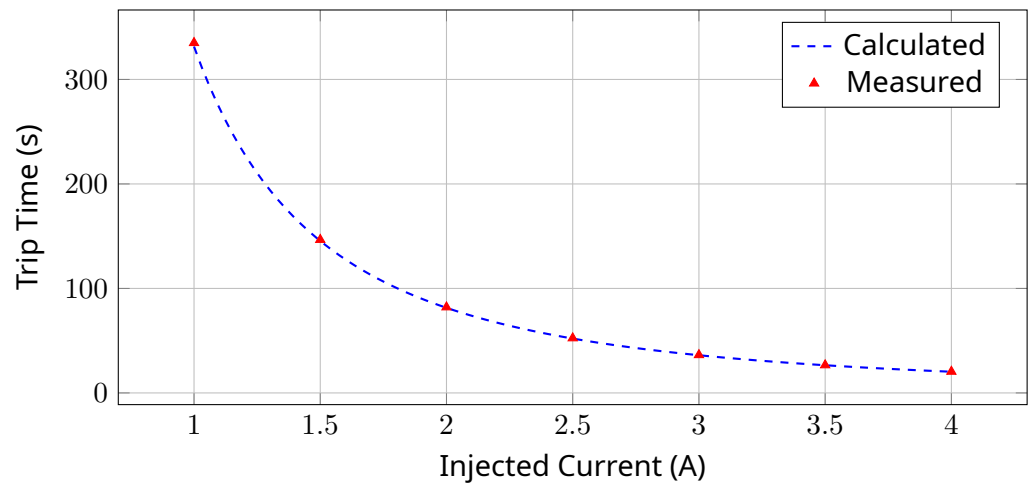


Figure 3: Results for Measurement 2.

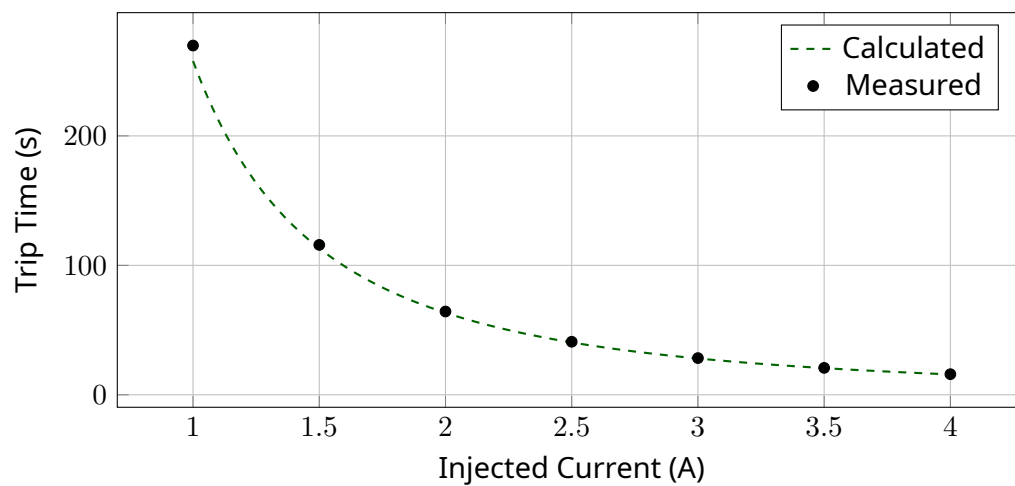
3. Measured vs Calculated (Measurement 3, $K=28$)

Figure 4: Results for Measurement 3.

4. Comparison: Measurement 1 vs Measurement 2

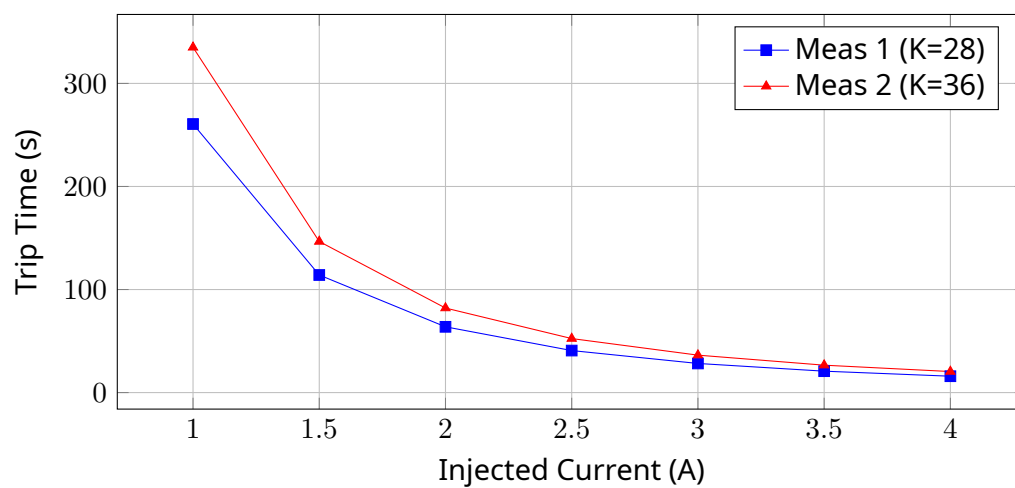


Figure 5: Comparing Measurements 1 and 2 shows how increasing the Machine Time constant (K) from 28 to 36 shifts the curve upwards, resulting in longer trip times for the same fault current.

5. Comparison: Measurement 1 vs Measurement 3

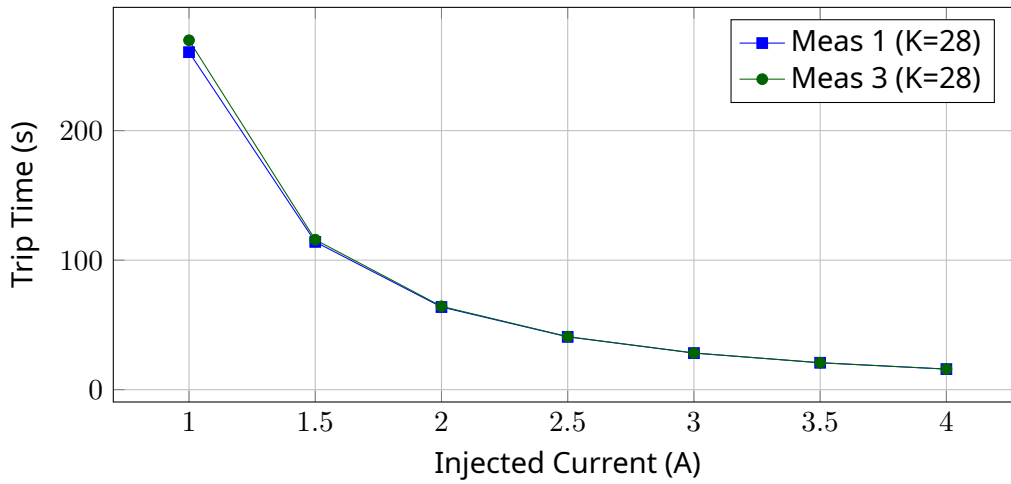


Figure 6: Comparing Measurement 1 and 3 (identical settings) demonstrates the consistency and repeatability of the protection relay.

5 Conclusion

The laboratory measurements successfully verified the protection logic against unbalanced loading. The Omicron test set was used to inject unbalanced currents into phases L2 and L3, effectively simulating a negative sequence condition where $I_2 \approx I_{injected}/3$.

The measured trip times aligned very closely with the theoretical values calculated using the formula $t = \frac{K}{(I_2/I_N)^2 - I_{start}^2}$.

- **Measurements 1 and 3** ($K = 28$) showed excellent repeatability, with only minor variations in trip times at low currents (e.g., 1A injection).
- **Measurement 2** ($K = 36$) confirmed that increasing the machine time constant K directly increases the trip time, allowing the machine to withstand the unbalance for a longer duration.